

A photograph of a nuclear power plant at night. Several large, hyperboloid cooling towers are visible, some with red lights around their upper sections. Tall smokestacks are also present, with thick white steam or smoke billowing from them into a dark, cloudy sky. In the foreground, there are industrial buildings, including one with prominent red piping or structures. The overall scene is illuminated by the plant's lights, creating a high-contrast image.

Energy Industry Projections

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Data Collection & Expected Signs of Variables

Dependent Variables:

- Yearly Global Fossil Fuel Energy Consumption (*TWh*)
- Yearly Global Renewable Energy Consumption (*TWh*)

Independent Variables:

- Yearly World GDP (*USD*) (+) (+)
- Yearly Crude Oil Prices (*USD*) (-) (+)
- Yearly Global Tertiary Education Enrollments (+) (+)
- Yearly Global Nuclear Powerplants Built (-) (+)
- Environmentalism Movement (*Dummy Variable for Year > 2007*) (-) (+)

Generate Log Differences of Variables

- Approximates growth rates
- Reduces heteroskedasticity
- Reduces multicollinearity
- Prevents trending relationships from inflating results (spurious correlations)

"How much did a variable change in percentage from one year to the next?"

Empirical Models

$$\begin{aligned} \text{dln_FossilFuelTotal}_t = & \beta_0 + \beta_1 * \text{dln_WorldGDP}_t + \beta_2 * \text{dln_CrudeOilPrice} + \beta_3 * \\ & \text{dln_TertiaryEnrollment}_t + \beta_4 * \text{dln_NuclearPlants} + \beta_5 * \text{EnviroDummy} + \varepsilon_t \end{aligned}$$

$$\begin{aligned} \text{dln_RenewableTotal}_t = & \beta_0 + \beta_1 * \text{dln_WorldGDP}_t + \beta_2 * \text{dln_CrudeOilPrice} + \beta_3 * \\ & \text{dln_TertiaryEnrollment}_t + \beta_4 * \text{dln_NuclearPlants} + \beta_5 * \text{EnviroDummy} + \varepsilon_t \end{aligned}$$

Predicting Fossil Fuel Energy Consumption

```
. reg dln_FossilFuelTotal dln_WorldGDP dln_CrudeOilPrice dln_TertiaryEnrollment dln_NuclearPlants EnviroDummy
```

Source	SS	df	MS	Number of obs	=	51
				F(5, 45)	=	6.24
Model	.008072011	5	.001614402	Prob > F	=	0.0002
Residual	.011643522	45	.000258745	R-squared	=	0.4094
				Adj R-squared	=	0.3438
Total	.019715533	50	.000394311	Root MSE	=	.01609

dln_FossilFuelTotal	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
dln_WorldGDP	.176529	.0486956	3.63	0.001	.078451	.2746071
dln_CrudeOilPrice	.004592	.0104722	0.44	0.663	-.0165001	.0256842
dln_TertiaryEnrollment	.0902082	.1268508	0.71	0.481	-.1652823	.3456988
dln_NuclearPlants	.0072385	.0034326	2.11	0.041	.0003248	.0141521
EnviroDummy	-.0022642	.0054171	-0.42	0.678	-.0131747	.0086463
_cons	.0047125	.0058947	0.80	0.428	-.00716	.016585

- Model explains 34.38% of variation within fossil fuel based energy consumption
- Statistically Significant Variables: WorldGDP & Nuclear Plants Built (+?)
- Surprising to see Crude Oil have a low impact and positive coefficient

Further Testing

Variance Inflation Factor Test

```
. vif
```

Variable	VIF	1/VIF
dln_CrudeOve	1.52	0.655787
dln_WorldGDP	1.47	0.678328
EnviroDummy	1.15	0.868132
dln_Nuclea~s	1.14	0.877222
dln_Tertia~t	1.08	0.927430
Mean VIF	1.27	

Heteroskedasticity Test

```
. estat hettest
```

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: Normal error terms
Variable: Fitted values of `dln_FossilFuelTotal`

H0: Constant variance

chi2(1) = 0.82
Prob > chi2 = 0.3644

Omitted Variable Test

```
. estat ovtest
```

Ramsey RESET test for omitted variables
Omitted: Powers of fitted values of `dln_FossilFuelTotal`

H0: Model has no omitted variables

F(3, 42) = 1.92
Prob > F = 0.1414

No signs of significant multicollinearity, heteroskedasticity, or omitted variables

Predicting Renewable Energy Consumption

```
. reg dln_RenewableTotal dln_WorldGDP dln_CrudeOilPrice dln_TertiaryEnrollment dln_NuclearPlants EnviroDummy
```

Source	SS	df	MS	Number of obs	=	51
Model	.003334542	5	.000666908	F(5, 45)	=	7.10
Residual	.004228574	45	.000093968	Prob > F	=	0.0001
Total	.007563115	50	.000151262	R-squared	=	0.4409
				Adj R-squared	=	0.3788
				Root MSE	=	.00969

dln_RenewableTotal	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
dln_WorldGDP	.0248489	.0293457	0.85	0.402	-.0342564	.0839542
dln_CrudeOilPrice	.0019446	.0063109	0.31	0.759	-.0107662	.0146555
dln_TertiaryEnrollment	-.384521	.0764448	-5.03	0.000	-.5384887	-.2305533
dln_NuclearPlants	.004658	.0020686	2.25	0.029	.0004916	.0088244
EnviroDummy	.0041742	.0032645	1.28	0.208	-.0024009	.0107492
_cons	.0298719	.0035523	8.41	0.000	.0227171	.0370267

- Marginally better results than the fossil fuel regression
- Model explains 37.88% of variation within renewable energy consumption
- Statistically Significant Variables: Tertiary Education Enrollments (-?) & Nuclear Plants Built

Further Testing (again)

Variance Inflation Factor Test

```
. vif
```

Variable	VIF	1/VIF
dln_CrudeOve	1.52	0.655787
dln_WorldGDP	1.47	0.678328
EnviroDummy	1.15	0.868132
dln_Nuclea~s	1.14	0.877222
dln_Tertia~t	1.08	0.927430
Mean VIF	1.27	

Heteroskedasticity Test

```
. estat hettest
```

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: Normal error terms
Variable: Fitted values of `dln_RenewableTotal`

H0: Constant variance

chi2(1) = 0.10
Prob > chi2 = 0.7498

Omitted Variable Test

```
. estat ovtest
```

Ramsey RESET test for omitted variables
Omitted: Powers of fitted values of `dln_RenewableTotal`

H0: Model has no omitted variables

F(3, 42) = 3.49
Prob > F = 0.0239

No signs of significant multicollinearity, heteroskedasticity, or omitted variables

Why does this matter to business majors?

```
. reg dln_ChevronStock dln_WorldGDP dln_CrudeOilPrice dln_TertiaryEnrollment dln_NuclearPlants EnviroDummy
```

Source	SS	df	MS	Number of obs	=	51
Model	.197109971	5	.039421994	F(5, 45)	=	1.59
Residual	1.11466381	45	.024770307	Prob > F	=	0.1819
Total	1.31177378	50	.026235476	R-squared	=	0.1503
				Adj R-squared	=	0.0558
				Root MSE	=	.15739

dln_ChevronStock	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
dln_WorldGDP	.6347096	.4764527	1.33	0.190	-.3249155	1.594335
dln_CrudeOilPrice	.1269868	.1024633	1.24	0.222	-.0793848	.3333585
dln_TertiaryEnrollment	.4721215	1.241146	0.38	0.705	-2.027675	2.971918
dln_NuclearPlants	.0058849	.0335859	0.18	0.862	-.0617606	.0735303
EnviroDummy	-.0025144	.0530021	-0.05	0.962	-.1092661	.1042374
_cons	.0610587	.0576753	1.06	0.295	-.0551054	.1772228

- Model explains 0.056% of variation within Chevron stock
- Statistically Significant Variables: None ☹️

Further Testing (again, again)

Variance Inflation Factor Test

```
. vif
```

Variable	VIF	1/VIF
dln_CrudeOve	1.52	0.655787
dln_WorldGDP	1.47	0.678328
EnviroDummy	1.15	0.868132
dln_Nuclea~s	1.14	0.877222
dln_Tertia~t	1.08	0.927430

Mean VIF	1.27	
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```
. estat hettest
```

Heteroskedasticity Test

```
. estat hettest
```

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: Normal error terms
Variable: Fitted values of `dln_ChevronStock`

H0: Constant variance

chi2(1) = 2.15
Prob > chi2 = 0.1427

Omitted Variable Test

```
. estat ovtest
```

Ramsey RESET test for omitted variables
Omitted: Powers of fitted values of `dln_ChevronStock`

H0: Model has no omitted variables

F(3, 42) = 3.11
Prob > F = 0.0362

No signs of significant multicollinearity, heteroskedasticity, or omitted variables

One final regression!

```
. reg dln_NEESock dln_WorldGDP dln_CrudeOilPrice dln_TertiaryEnrollment dln_NuclearPlants EnviroDummy
```

Source	SS	df	MS	Number of obs	=	36
Model	.0393785	5	.0078757	F(5, 30)	=	0.51
Residual	.45886869	30	.015295623	Prob > F	=	0.7628
				R-squared	=	0.0790
				Adj R-squared	=	-0.0745
Total	.49824719	35	.014235634	Root MSE	=	.12368

dln_NEESock	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
dln_WorldGDP	.2803077	.4825143	0.58	0.566	-.7051179	1.265733
dln_CrudeOilPrice	-.1064215	.0879279	-1.21	0.236	-.2859942	.0731513
dln_TertiaryEnrollment	.7911505	1.315831	0.60	0.552	-1.896135	3.478436
dln_NuclearPlants	-.0067154	.0295641	-0.23	0.822	-.0670934	.0536626
EnviroDummy	.0449871	.0456695	0.99	0.332	-.0482824	.1382567
_cons	.0685848	.064979	1.06	0.300	-.06412	.2012897

- Model explains... less than nothing?
- Statistically Significant Variables: None ☹️

Further Testing (again, again, again)

Variance Inflation Factor Test

```
. vif
```

Variable	VIF	1/VIF
dln_CrudeO~e	1.46	0.684868
dln_WorldGDP	1.39	0.717334
dln_Nuclea~s	1.22	0.817815
EnviroDummy	1.17	0.857169
dln_Tertia~t	1.13	0.885070
Mean VIF	1.27	

```
. estat hettest
```

Heteroskedasticity Test

```
. estat hettest
```

Breusch-Pagan/Cook-Weisberg test for heteroskedasticity
Assumption: Normal error terms
Variable: Fitted values of `dln_NEESTock`

H0: Constant variance

chi2(1) = 0.07
Prob > chi2 = 0.7909

Omitted Variable Test

```
. estat ovtest
```

Ramsey RESET test for omitted variables
Omitted: Powers of fitted values of `dln_NEESTock`

H0: Model has no omitted variables

F(3, 27) = 0.06
Prob > F = 0.9810

No signs of significant multicollinearity, heteroskedasticity, or omitted variables

Thank you!

Data Source Citations

- <https://www.macrotrends.net/stocks/charts/NEE/nextera-energy/stock-price-history>
- <https://www.macrotrends.net/stocks/charts/CVX/chevron/stock-price-history>
- <https://inflationdata.com/articles/inflation-adjusted-prices/historical-crude-oil-prices-table/>
- <https://globalenergymonitor.org/projects/global-nuclear-power-tracker>
- <https://ourworldindata.org/fossil-fuels>
- <https://ourworldindata.org/renewable-energy>
- <https://ourworldindata.org/global-education>
- <https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?end=2023&start=1960>